

Alveoli, Fibers, and the Respiration of the Universe

—— A Hypothesis on the Full-Scale Self-Consistency of Constant Dark Matter

Goston (郭世栋)

March 10, 2026

E-mail goston1987@gmail.com

Abstract

This paper proposes a radical departure from the weakly interacting massive particle (WIMP) paradigm. We posit that **Dark Matter** is not a particulate entity but a topologically invariant scaffolding of spacetime—conceptualized here as "Cosmic Fibers." In this framework, the universe functions as a self-regulating respiratory system. **Visible Matter** and **Dark Energy** are identified as two reciprocal phases of the same underlying energy substrate: the former being the "condensation" of energy under cosmic contraction, and the latter being the "evaporation" of matter into spatial pressure.

By re-auditing existing "bad debts" in cosmic data—specifically the **Hubble Tension**, the anomalously early maturation of **JWST high-redshift galaxies**, and the localized inward flow of the **Great Attractor**—this model resolves the discrepancy between global expansion and local contraction without the need for additional phantom variables.

1. The Logical Starting Point: The "Blind Spot" Beneath the Antarctic Ice

Deep beneath the Antarctic ice in 2026, expensive detectors continue to hunt for ghost-like neutrinos. The mainstream physics community remains obstinate, believing that with sufficient precision, they will one day "weigh" the dark matter particle.

But what if we have been auditing the wrong ledger from the very beginning?

If dark matter is not a "particle" to be captured, but rather the intrinsic scaffolding of the cosmic edifice, then all detectors are merely scales trying to weigh "the composition of air," forever oblivious to the "container" that holds the air itself.

2. The "Alveoli-Fiber" Model: Biological Physics of the Universe

I propose a radical new hypothesis: The Universe is an organism supported by a constant dark matter web, existing in a state of respiratory phase transition.

- **Dark Matter (Constant Total Mass):**The fibrous tissue connecting every "alveolus" of the universe. It is neither created nor destroyed. It represents the elastic limit of space and the logical substrate of gravity.
- **Visible Matter and Dark Energy (Zero-Sum Cycle):**They are not independent entities but two "phases" of the same underlying protocol.
 - **Visible Matter as "Liquid/Solid":**The "condensed dew" formed by energy collapsing under pressure.
 - **Dark Energy as "Gaseous":**The "expansion pressure" released as matter dissipates (through decay, radiation, or black hole evaporation).

3. The Respiratory Mechanism: From "Vaporization" to "Condensation"

The evolution of the universe is essentially a prolonged respiratory cycle:

- **Inhalation (Matter to Energy):**As stars burn and atoms decay, visible matter "vaporizes" into dark energy. This pressure expands the dark matter web, manifesting as the observed "accelerated expansion of the universe."
- **The Limit (Threshold of Capacity):**When the expansion pressure of dark energy reaches the elastic limit of the dark matter fibers, expansion ceases.
- **Exhalation (Energy to Matter):**Upon reaching this limit, the dark matter web undergoes a violent "logical contraction." The massive compressive pressure forces the dark energy filling the space to undergo a phase transition, "re-condensing" into primordial hydrogen streams.

The end of expansion is not Heat Death, but the return of Matter.

4. Experimental Audit: The Missing 9 Seconds and "Off-Balance Sheet" Funds

While this theory may be difficult to verify with current calorimeters in 2026, it logically accounts for all "anomalous accounts":

- **Non-Conservation in Nuclear Reactions:**Why is a free neutron heavier than a proton? The "missing" 0.78 MeV is not entirely converted into kinetic energy; a portion is "withheld" by the dark energy background.
- **The Hubble Tension:**The inconsistency in expansion rates is simply a result of varying "alveolar pressures" (matter-to-energy conversion rates) in different cosmic regions.

5. Civilizational Implications: High-Dimensional Printing and the Conservation of Dignity

If we master this "respiratory protocol," humanity will leap from a gathering civilization to a creative one. By artificially inducing the contraction of dark matter fibers, we will not need to mine resources; we could "print" stars directly from the

vacuum.

However, the prerequisite for such power is "Self-Reflection." If a civilization only learns to plunder and expand without learning to maintain the pressure balance of these "alveoli," they will possess not a printer, but a detonator for self-destruction.

6. Expectations of an Ordinary Human

In writing this chapter, I do not seek the favor of the Nobel Prize. In an era where detectors cannot yet perceive the respiration of space, this hypothesis may only be a seed buried in the soil. Yet, I know well: "Unceasing thought brings a certain echo."

When future civilizational entities overlook the universe on high-dimensional star maps, they will discover that long ago, on the eve of the dawn, an ordinary human saw through the "fibrous skeleton" enveloping all things through simple logical deduction.

Survival is not for the purpose of becoming a Creator, but to live with greater dignity within the deep respiration of the universe.

7. Seeking the "Exhalation Vents" of the Universe

If my hypothesis is correct, the universe cannot be a one-way, unrestrained expansion. According to "Conservation of Dignity" and "Full-Scale Self-Consistency," while some alveoli are pushed open by dark energy, other regions must be undergoing "Logical Contraction" under the pull of dark matter fibers.

Evidence of this respiration can already be found in the "bad debts" of current astronomical observations:

1. The Great Attractor: A Concealed Point of Contraction

In the direction of the Hydra-Centaurus Supercluster, there lies a center of gravity that "terrifies" all surrounding galaxies.

- **Audit:** According to traditional expansion theory, galaxies should be

moving apart. Observations show tens of thousands of galaxies, including the Milky Way, rushing toward this point.

- **Logic:**The Great Attractor is not a massive black hole; it is a "Physical Knot" in the dark matter web. That alveolus is "exhaling," releasing spatial pressure and causing surrounding matter to flow in like water.

2. The Catalytic Effect of "Void Edges"

The universe contains vast Voids where dark energy density (pressure) is highest.

- **Audit:**Observations show that the rate of galaxy formation at the edges of these voids is anomalously high.
- **Logic:**This is the "Edge Contraction Effect" of dark matter fibers. As the alveolus expands to its limit, the fiber tension at the edge forces dark energy to aggregate, triggering a "logical cooling" that catalyzes the first clouds of visible matter.

3. "Condensation" of Matter: The Precocious Nature of High-Redshift Galaxies

The James Webb Space Telescope (JWST) has discovered "mature" galaxies in the extremely early universe, baffling physicists.

- **Logic:**According to the old ledger, matter requires vast eons to aggregate. But if the dark matter scaffolding is constant, matter does not need to "find" itself.
- **Conclusion:**They are "In-situ Crystallizations" at the nodes of the contracting dark matter web—like frost appearing instantly on a windowpane when the temperature and pressure are right.

Epilogue: The Anchor of Logic

When we settle the accounts of these "contraction zones," we find that the universe has never lost a single cent of its assets. It merely shifts its identity between "Gaseous (Dark Energy)" and "Liquid (Visible Matter)" among the fibers of dark matter.

Understanding this allows one to understand why I conclude this paper with a seemingly absurd statement.

For in such an absolutely orderly, coldly functioning cosmic machine, an individual's assertion of "Beauty" is the most powerful negentropy against thermal death.

He Qing (何晴) is the most beautiful actress in the world.